



HealthConnect No-Show Model - Specification for ML Engineering Handoff

Week 8 - Data Science Track

AnalystLab Africa - Experience Lab Internship Programme

This document is the full model specification referenced in the handoff message sent to ML Engineering. It describes the model as it exists today, and explicitly separates what is ready for integration from what must change before deployment per ML Engineering's Week 8 review feedback.

1. Model Overview

Property	Value
Algorithm	Random Forest Classifier (scikit-learn)
Hyperparameters	max_depth = 6, min_samples_leaf = 20
Target	no_show (1 = patient did not attend, 0 = attended)
Training data	HealthConnect_Appointment_Data.csv (synthetic dataset, approved project resource)
Validated performance	ROC-AUC $\approx$ 0.682 (time-based split), $\approx$ 0.669 (patient-level split, unseen patients)
Recommended operating threshold	0.30–0.35 (not the scikit-learn default of 0.5)

2. Input Features - Current State (Post-Fix)

✓ waiting_time_minutes has been removed and the model retrained and re-verified (see Section 3). The feature list below is the corrected, final feature set.

2.1 Numeric Features

Feature	Type	Description
age	int	Patient age in years
booking_lead_days	int	Days between booking and appointment date
previous_appointments	int	Total prior appointments for this patient
previous_no_shows	int	Total prior no-shows for this patient
distance_to_clinic_km	float	Distance from patient's address to clinic
no_show_rate_history	float (derived)	previous_no_shows / previous_appointments (0 if no prior appointments)
is_new_patient	int, 0/1 (derived)	1 if previous_appointments = 0
long_lead_time	int, 0/1 (derived)	1 if booking_lead_days above the 75th percentile
reminder_received	int, 0/1 (derived)	1 if reminder_sent = 'Yes'
is_weekend_appointment	int, 0/1 (derived)	1 if appointment_day is Saturday or Sunday
mid_lead_time	int, 0/1 (derived)	1 if booking_lead_days between 10 and 30
distance_lead_interaction	float (derived)	distance_to_clinic_km $\times$ booking_lead_days
high_risk_history	int, 0/1 (derived)	1 if no_show_rate_history > 0.3
prevnoshow_x_leaddays	int (derived)	previous_no_shows $\times$ booking_lead_days

2.2 Categorical Features (one-hot encoded before training)

Feature	Example Values
gender	Female / Male / Prefer not to say
appointment_type	Follow-up / Specialist Consultation / etc.
appointment_day	Monday–Sunday
appointment_time	Morning / Afternoon / Evening
reminder_channel	SMS / WhatsApp / Email / None
lead_band	0-7 / 8-14 / 15-30 / 31-60 (derived from booking_lead_days)

Note on encoding: the notebook currently encodes these with pandas `get_dummies()` applied separately from the model object, not inside a scikit-learn Pipeline. This must be repackaged (see Section 4) before the model can accept raw string categoricals directly, as ML Engineering's pipeline requires.

3. Data Leakage Issue - Identified and Resolved

ML Engineering's handoff review identified that `waiting_time_minutes` cannot be known at prediction time for an appointment that has not yet occurred. This was fixed this week: the model was retrained without this feature and re-verified end-to-end on the real HealthConnect dataset.

Result: ROC-AUC 0.682 (identical to 3 decimal places with or without the feature), confirming it carried negligible real signal. The lead-time blind spot and gender fairness gap are both unaffected. The one substantive change: the baseline comparison is now more conservative, the previously-reported statistically significant improvement over the Week 5 baseline no longer holds once the leakage feature is correctly removed from both models (bootstrap 95% CI [-0.0019, 0.0146], includes zero). This is reported transparently as a revision to the Week 7 framing.

4. Technical Constraints - Status

Constraint (from ML Engineering)	Current Status
Pickle-compatible scikit-learn object	Model itself is a plain <code>RandomForestClassifier</code> (pickleable). Not yet packaged with preprocessing, see below.
Categorical features as raw strings	Not yet supported, current notebook code applies <code>pd.get_dummies()</code> manually outside the model object. Needs a scikit-learn Pipeline (e.g. <code>ColumnTransformer</code> + <code>OneHotEncoder</code>) trained and pickled together with the classifier so raw strings can be passed in directly.
<code>waiting_time_minutes</code> excluded	✅ Done - retrained and re-verified this week. See Section 3.

5. Recommended Output Format

Mapping proposed for ML Engineering's required output schema, based on the 0.30–0.35 recommended threshold:

no_show_probability range	risk_tier	recommended_action
< 0.30	LOW RISK	NO ACTION
0.30 – 0.60	MODERATE RISK	TRIGGER_SMS_REMINDER_REQUESTING_CONFIRMATION
> 0.60	HIGH RISK	TRIGGER_PHONE_CALL_CONFIRMATION

This tiering is a proposed starting point based on the threshold analysis in Week 7 (Part 3) , it has not yet been separately validated at the tier level and should be reviewed once real usage data is available.

6. Known Model Limitations (carried over from Week 7–8 testing)

- Lead-time blind spot: the model misses ~91–100% of no-shows booked within 14 days of the appointment, versus only 1.0% for the 31–60 day band. This is the model's most significant limitation and should be communicated to any team building on top of these predictions.

- Gender fairness gap: ROC-AUC 0.723 (male) vs. 0.641 (female); an attempted fix (interaction features) did not close this gap. Reported as unresolved.
- Synthetic training data: all figures above need revalidation once real HealthConnect data is available.

7. Summary - What's Ready vs. What's Pending

Item	Status
Model logic and evaluation	Ready - thoroughly tested in Week 7–8
Recommended threshold	Ready - 0.30–0.35
Output tiering proposal	Ready - pending validation once live
Feature set (waiting_time_minutes removed)	✅ Ready - fixed and re-verified this week
Baseline significance claim	Revised - no longer statistically significant; reported honestly
Packaged pipeline (raw strings in, pickle out)	Pending - still required before handoff
Fairness gap	Open - documented, not resolved